

INDIA_CRYPTO_SOC

AES-256-CA · PDF Encryption & Decryption Proof

Round-trip verification: original PDF → encrypt → decrypt → identical PDF

Overview

Parameter	Value
Algorithm	AES-256-CA (CTR mode, 14 rounds × 4 CA stages = 56 CA ops/block)
Key size	256 bits (32 bytes)
IV / Nonce	128 bits (16 bytes)
Input file	INDIA_CRYPTO_SOC_Demo.pdf
Input size	89,755 bytes (87.7 KB)
Output (enc)	89,755 bytes (same size — CTR mode is length-preserving)
Bytes changed	89,419 / 89,755 (99% diffusion)
Round-trip	■ PASS — decrypted == original (byte-for-byte)

Key Material

Field	Hex Value (256-bit AES-256 key / 128-bit IV)
KEY	de ad be ef ca fe ba be 12 34 56 78 9a bc de f0 0f ed cb a9 87 65 43 21 aa bb cc dd ee ff 00 11
IV	01 23 45 67 89 ab cd ef fe dc ba 98 76 54 32 10

SHA-256 Hash Verification

A byte-for-byte match is proven when the SHA-256 of the decrypted output exactly equals the SHA-256 of the original PDF. Any single corrupted bit would produce a completely different hash.

File / Stage	SHA-256 (hex)
Original PDF	1e8cc58ff3e7d5d70cd001a973d06f9cdf96683d1ee21000c66c2e3da26d1102
Ciphertext (enc)	5e306d281f10c03048d522de961b00915546f9af2643c58540ed26b2696b7607
Decrypted PDF	1e8cc58ff3e7d5d70cd001a973d06f9cdf96683d1ee21000c66c2e3da26d1102
Match (orig ↔ dec)	■ IDENTICAL

Byte-Level Evidence

The table below shows the first 32 bytes of the original PDF, after encryption, and after decryption — confirming the ciphertext is completely transformed and the decrypted output is identical to the original.

Stage	First 32 bytes (hex)																																							
Original PDF	25	50	44	46	2d	31	2e	34	0a	25	93	8c	8b	9e	20	52	65	70	6f	72	74	4c	61	62	20	47	65	6e	65	72	61	74								
Encrypted	bd	ea	fe	0c	b9	a6	3b	58	52	7d	73	3f	01	0c	32	63	5a	1e	15	89	e4	aa	eb	a8	fa	aa	6e	6a	81	df	6b	3b								
Decrypted PDF	25	50	44	46	2d	31	2e	34	0a	25	93	8c	8b	9e	20	52	65	70	6f	72	74	4c	61	62	20	47	65	6e	65	72	61	74								

Original PDF header (ASCII): %PDF-1.4 — confirms the decrypted output is a valid PDF (starts with %PDF).

HMAC-SHA256 Authentication Tag

The HMAC-SHA256 tag below authenticates the ciphertext, matching the CA-HMAC engine modeled in rtl/periph/india_pdf_engine.v. A recipient recomputes this tag before decryption; any modification to the ciphertext invalidates the tag.

54521af5233279d58e5084abe2846807d6d5c7dacf4c7d049bbadc9bd233feb0

■ **ROUND-TRIP VERIFIED — The decrypted PDF is byte-for-byte identical to the original**

AES-256-CA (CTR mode) is a symmetric cipher: the same key + IV decrypts what was encrypted. The decrypted file passes the PDF validity check (%PDF header intact) and its SHA-256 matches the original exactly.